

Mathematics for Mechatronics Engineering

Complex Numbers and Partial Fractions

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1st Semester 2026

- **Session 1:** Complex Numbers
- **Session 2:** Partial Fractions

Goal: Develop engineering intuition and practical problem-solving skills.

Why Engineers Use Complex Numbers

Complex numbers appear everywhere in engineering.

Engineering Applications

- AC circuits
- Electrical impedance
- Oscillations and vibrations
- Signal processing
- Control systems

Complex numbers simplify oscillation and rotation calculations.

Definition of Complex Numbers

A complex number has the form:

$$z = a + jb$$

where:

- a = real part
- b = imaginary part
- $j = \sqrt{-1}$

Engineering Convention

Engineers use j instead of i because i often represents current.

Complex numbers can be represented graphically.

$$z = 3 + j4$$

- Horizontal axis: Real part
- Vertical axis: Imaginary part

Interpretation

A complex number behaves like a vector in 2D space.

Addition and Subtraction

Example:

$$(2 + j3) + (1 - j2)$$

Combine real and imaginary parts separately:

$$= (2 + 1) + j(3 - 2)$$

$$= 3 + j$$

Physical Meaning

Complex addition behaves like vector addition.

Complex Multiplication

Recall:

$$j^2 = -1$$

Example:

$$(2 + j)(3 + 2j)$$

Expand:

$$= 6 + 4j + 3j + 2j^2$$

$$= 6 + 7j - 2$$

$$= 4 + 7j$$

Complex Conjugate

The conjugate of:

$$z = a + jb$$

is:

$$z^* = a - jb$$

Example:

$$(3 + j4)^* = 3 - j4$$

Engineering Application

Complex conjugates are used in:

- AC power calculations
- Communication systems

Magnitude of a Complex Number

Magnitude measures vector length.

$$|z| = \sqrt{a^2 + b^2}$$

Example:

$$z = 3 + j4$$

$$|z| = \sqrt{3^2 + 4^2} = 5$$

Interpretation

Magnitude represents amplitude or signal strength.

Polar Form

A complex number can also be written as:

$$z = r\angle\theta$$

where:

- r = magnitude
- θ = angle

Example:

$$3 + j4$$

$$r = 5$$

$$\theta = \tan^{-1}\left(\frac{4}{3}\right) = 53.1^\circ$$

Therefore:

Multiplication in Polar Form

Suppose:

$$z_1 = r_1 \angle \theta_1$$

$$z_2 = r_2 \angle \theta_2$$

Then:

$$z_1 z_2 = (r_1 r_2) \angle (\theta_1 + \theta_2)$$

Important Insight

Complex multiplication performs:

- Scaling
- Rotation

Euler's Formula

One of the most important formulas in engineering:

$$e^{j\theta} = \cos \theta + j \sin \theta$$

Why It Matters

Euler's formula connects:

- Exponentials
- Rotation
- Oscillation

Used heavily in AC analysis and control systems.

Electrical impedance:

$$Z = R + jX$$

where:

- R = resistance
- X = reactance

Key Idea

Complex numbers simplify AC circuit calculations.

Practice Exercise

Convert the following to polar form:

$$z = 1 + j1$$

Find:

- ① Magnitude
- ② Angle

Then express as:

$$r\angle\theta$$

Why Partial Fractions Matter

Partial fractions are widely used in engineering.

Applications

- Laplace transforms
- Control systems
- Circuit transient analysis
- Dynamic systems

They help simplify complicated rational functions.

Basic Idea of Partial Fractions

We break complicated fractions into simpler pieces.

Example:

$$\frac{5}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

Goal

Simpler fractions are easier to integrate and inverse-transform.

Distinct Linear Factors

Example:

$$\frac{3x + 5}{(x + 1)(x + 2)} = \frac{A}{x + 1} + \frac{B}{x + 2}$$

Multiply both sides:

$$3x + 5 = A(x + 2) + B(x + 1)$$

Solve for A and B .

Solving for Constants

Expand:

$$3x + 5 = Ax + 2A + Bx + B$$

Group terms:

$$3x + 5 = (A + B)x + (2A + B)$$

Match coefficients:

$$A + B = 3$$

$$2A + B = 5$$

Solution:

$$A = 2, \quad B = 1$$

Therefore:

$$\frac{3x + 5}{(x + 1)(x + 2)} = \frac{2}{x + 1} + \frac{1}{x + 2}$$

Important

The original complicated fraction becomes two simpler fractions.

Repeated Factors

Example:

$$\frac{2x + 1}{(x + 1)^2}$$

Use:

$$\frac{A}{x + 1} + \frac{B}{(x + 1)^2}$$

Rule

Repeated denominator terms require repeated partial fraction terms.

Quadratic Factors

Example:

$$\frac{2x + 3}{(x + 1)(x^2 + 1)}$$

Use:

$$\frac{A}{x + 1} + \frac{Bx + C}{x^2 + 1}$$

Important Rule

Irreducible quadratic factors require linear numerators.

Connection to Laplace Transforms

Suppose:

$$F(s) = \frac{5}{(s+1)(s+2)}$$

Partial fractions:

$$F(s) = \frac{5}{s+1} - \frac{5}{s+2}$$

Inverse Laplace transform:

$$f(t) = 5e^{-t} - 5e^{-2t}$$

Engineering Importance

This is heavily used in control systems and transient analysis.

Dynamic systems often produce transfer functions such as:

$$G(s) = \frac{1}{(s+1)(s+3)}$$

Partial fractions simplify system analysis and time-domain solutions.

Practice Exercise

Decompose:

$$\frac{4x + 1}{(x + 1)(x + 3)}$$

into partial fractions.

Find:

$$A, B$$

such that:

$$\frac{4x + 1}{(x + 1)(x + 3)} = \frac{A}{x + 1} + \frac{B}{x + 3}$$

- Complex numbers describe oscillation and rotation
- Polar form simplifies multiplication
- Euler's formula connects trigonometry and exponentials
- Partial fractions simplify rational functions
- Partial fractions are essential for inverse Laplace transforms